



RASD

Requirements Analysis and Specification Document

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Introduction

1.1 Purpose

1.1.1 General purpose

ROAd, Robotaxi Optimized Allocation, is a software system created to support the management of an autonomous taxi fleet within an urban environment. Its purpose is to help fleet operators keep track of robotaxis, manage passenger ride requests, assign vehicles effectively, and redistribute idle vehicles across the city based on expected demand.

The main goal of ROAd is to reduce passenger waiting times while improving the overall efficiency of the fleet. To achieve this, the system supports different vehicle allocation strategies, such as assigning the nearest available robotaxi or choosing the vehicle with the lowest estimated arrival time. The fleet operator can change the selected strategy at runtime, allowing the system to adapt to different operational needs. In addition, dynamic fleet rebalancing supports this process by moving idle taxis toward areas where demand is expected to be higher, using both historical and real-time data.

ROAd also supports advance ride booking, which requires the system to manage vehicle availability over time and avoid conflicts between immediate rides and scheduled bookings. The system explicitly follows the lifecycle of each robotaxi, from availability to assignment, arrival, ride execution, and maintenance. Passengers are informed whenever relevant changes occur in the state of their assigned vehicle.

The project focuses on the software architecture and management logic of the robotaxi allocation platform. Autonomous driving and payment processing are outside the project scope. Mapping and routing information is obtained through external services.

1.1.2 Goals

- ▶ **[G1]** Users must be able to log in, and manage their personal account. Also, passengers must be able to register.
- ▶ **[G2]** Passengers must be able to request an immediate robotaxi ride by providing pickup and destination locations.
- ▶ **[G3]** Passengers must be able to book a robotaxi ride in advance by specifying pickup location, destination, date, and time.
- ▶ **[G4]** The system must be able to assign an available robotaxi to a ride request according to the globally activate allocation strategy.

- ▶ **[G5]** The fleet operator must be able to select and change the vehicle allocation strategy at runtime.
- ▶ **[G6]** The system must be able to track and update the state of each robotaxi during its lifecycle.
- ▶ **[G7]** Passengers must be notified about relevant updates concerning their ride, such as vehicle assignment, vehicle arrival, and ride status changes.
- ▶ **[G8]** The fleet operator must be able to monitor the current status and position of the robotaxi fleet.
- ▶ **[G9]** The system must be able to identify zones with expected high demand and automatically trigger the rebalancing of idle robotaxis.
- ▶ **[G10]** The system must prevent conflicting assignments or reservations of the same robotaxi for overlapping rides or bookings.

1.2 Scope

1.2.1 General Scope

The S2B will interact with both passengers and fleet operators. For passengers, it will provide a mobile-friendly web application that allows them to request or book robotaxi rides. For fleet operators, it will provide a web-based graphical user interface to monitor the autonomous fleet.

The main functionality offered to passengers is the possibility to request a robotaxi ride directly from the application, either immediately or in advance, making urban transportation simpler and helping reduce waiting times.

Fleet operators are responsible for supervising the robotaxi fleet, selecting the vehicle allocation strategy, monitoring vehicle states and positions. The system also supports demand-driven automatic rebalancing, improving fleet efficiency and reducing passenger waiting times.

1.2.2 Phenomena

The ROAd system includes several interactions among passengers, fleet operators, robotaxis, and external services.

Phenomenon	Controller	Shared
Passenger decides to use the ROAd service	W	N
Passenger registration	W	Y
Passenger login	W	Y
Passenger submits an immediate ride request	W	Y
Passenger submits an advance ride booking request	W	Y
Passenger provides pickup and destination locations	W	Y
Passenger cancels a pending or scheduled ride	W	Y
Fleet operator login	W	Y
Fleet operator selects the vehicle allocation strategy	W	Y
Fleet operator monitors fleet status and positions	W	Y
Robotaxi sends its current position to the system	W	Y

Phenomenon	Controller	Shared
Robotaxi sends its current operational state to the system	W	Y
Robotaxi is moved toward a rebalancing zone	W	Y
Robotaxi enters maintenance	W	Y
External mapping service provides route and travel-time information	W	Y
External demand data source provides historical or real-time demand data	W	Y
Ride starts when the passenger boards the assigned robotaxi	W	Y
Ride ends when the passenger reaches the destination	W	Y
System validates user credentials	M	N
System stores passenger and fleet operator account data	M	N
System checks robotaxi availability	M	N
System checks for conflicts between immediate rides and advance bookings	M	N
System assigns a robotaxi to a ride request	M	Y
System reserves a robotaxi for an advance booking	M	Y
System applies the currently selected allocation strategy	M	N
System updates the state of a robotaxi	M	Y
System updates the state of a ride	M	Y
System marks a robotaxi as unavailable during maintenance	M	Y
System notifies the passenger about vehicle assignment	M	Y
System notifies the passenger about robotaxi arrival	M	Y
System notifies the passenger about ride status changes	M	Y
System identifies zones with expected high demand	M	N
System automatically triggers the rebalancing of idle robotaxis	M	Y

1.3 Definitions, Acronyms, Abbreviations

1.3.1 Definitions

- **User:** a generic actor who interacts with the ROAd system. Depending on the context, a user can be either a passenger or a fleet operator.

- ▶ **Passenger:** a person who uses the ROAd mobile-friendly web application to request an immediate robotaxi ride, book a ride in advance, monitor the assigned vehicle, and receive ride-related notifications.
- ▶ **Fleet Operator:** a person responsible for supervising and managing the robotaxi fleet through the ROAd web interface. The fleet operator can monitor vehicles, select allocation strategies, and observes fleet rebalancing.
- ▶ **Robotaxi:** an autonomous taxi belonging to the fleet managed by ROAd. A robotaxi can be assigned to passengers, execute rides, become unavailable, or enter maintenance.
- ▶ **Fleet:** the set of robotaxis managed by the ROAd system in the considered urban area.
- ▶ **Ride Request:** a request submitted by a passenger to travel from a pickup location to a destination. A ride request can be immediate or scheduled in advance.
- ▶ **Immediate Ride:** a ride requested by a passenger to be served as soon as possible.
- ▶ **Advance Booking:** a ride request scheduled by a passenger for a future date and time. The system must reserve a suitable robotaxi while avoiding conflicts with other assignments or reservations.
- ▶ **Pickup Location:** the location where the passenger wants to be picked up by the assigned robotaxi.
- ▶ **Destination:** the location where the passenger wants to be dropped off.
- ▶ **Allocation Strategy:** a policy used by the system to select which available robotaxi should be assigned to a ride request. Examples include selecting the nearest available vehicle or the vehicle with the minimum estimated arrival time.
- ▶ **Runtime Strategy Selection:** the possibility for the fleet operator to change the active allocation strategy while the system is running, without stopping the service.
- ▶ **Vehicle State:** the current lifecycle condition of a robotaxi. In ROAd, relevant states include available, rebalancing, assigned, arriving, arrived, in ride, and maintenance.
- ▶ **Available Robotaxi:** a robotaxi that is currently not assigned to any ride, not executing a ride, and not under maintenance.
- ▶ **Assigned Robotaxi:** a robotaxi that has been selected by the system to serve a passenger's ride request.
- ▶ **Idle Robotaxi:** a robotaxi that is available and not currently serving a passenger. Idle robotaxis may be considered for fleet rebalancing.
- ▶ **Fleet Rebalancing:** the process of repositioning idle robotaxis toward areas where high demand is expected, in order to reduce future passenger waiting times.
- ▶ **Demand Zone:** an area of the city associated with expected or detected ride demand.
- ▶ **High-Demand Zone:** a demand zone where the system predicts or detects a high number of ride requests.
- ▶ **Estimated Time of Arrival:** the estimated time needed by a robotaxi to reach a passenger's pickup location.

- ▶ **Notification:** a message sent by the system to a passenger or fleet operator to communicate relevant events, such as vehicle assignment, vehicle arrival, ride status changes, or rebalancing information.
- ▶ **Maintenance:** a condition in which a robotaxi is unavailable because it requires inspection, repair, cleaning, or any other operation that prevents it from serving rides.
- ▶ **External Mapping Service:** an external service used by ROAd to obtain geographical information, routes, distances, and travel-time estimations.
- ▶ **External Demand Data Source:** an external or historical data source used by ROAd to estimate future ride demand in different city zones.
- ▶ **Conflict:** a situation in which the same robotaxi would be assigned or reserved for two or more overlapping rides or bookings. The system must prevent such situations.

1.3.2 Acronyms

- ▶ **ROAd:** Robotaxi Optimized Allocation.
- ▶ **S2B:** Software To Be, the software system described and specified in this document.
- ▶ **API:** Application Programming Interface.
- ▶ **ETA:** Estimated Time of Arrival.
- ▶ **GPS:** Global Positioning System.
- ▶ **UI:** User Interface.
- ▶ **GUI:** Graphical User Interface.
- ▶ **FSM:** Finite State Machine.
- ▶ **UML:** Unified Modeling Language.

1.3.3 Abbreviations

- ▶ **[Gn]:** used to identify the n-th goal.
- ▶ **[Rn]:** used to identify the n-th functional requirement.
- ▶ **[NFRn]:** used to identify the n-th non-functional requirement.
- ▶ **[Dn]:** used to identify the n-th domain assumption.
- ▶ **[FSMn]:** used to identify the n-th finite state machine.

1.4 Document Structure

- ▶ **Chapter 1 - Introduction:** this chapter introduces the ROAd system, describing its purpose, goals, scope, main phenomena, and terminology. It also provides the structure of the document.
- ▶ **Chapter 2 - Overall Description:** this chapter provides a general description of the system from a requirements perspective. It includes relevant scenarios, the domain model, user characteristics, main product functions, non-functional aspects, and the main assumptions, dependencies, and constraints.

- ▶ **Chapter 3 - Additional Models:** this chapter provides additional models used to clarify the expected behavior of the system. In particular, it includes requirements-level sequence diagrams and finite state machines for the most relevant system processes.
- ▶ **Chapter 4 - References:** this chapter lists the documents, papers, standards, and other sources used as references for the preparation of this document.

Overall Description

2.1 Scenarios

This section provides an informal description of relevant scenarios to illustrate the shared phenomena between the ROAd system and the external world, highlighting the interactions involving the main actors: Passengers and Fleet Operators.

Scenario 1: Immediate Ride Request and Vehicle Lifecycle

Alice, a registered Passenger, needs to travel from her office to the city center. She opens the ROAd mobile-friendly web application and submits an immediate ride request by providing her pickup location and destination. The system processes the request and assigns a robotaxi according to the currently active allocation strategy. The chosen robotaxi passes from the available state to the assigned state, and Alice receives a notification on her app confirming the assignment. Shortly after, the robotaxi starts moving towards Alice's pickup location, transiting to the arriving state, and Alice is notified again. Once the vehicle reaches the pickup location, it enters the arrived state, and Alice receives a final notification to board. Alice boards the robotaxi, changing its state to in-ride. Upon reaching the destination, Alice exits the vehicle, and the robotaxi becomes available again to serve new requests.

Scenario 2: Advance Ride Booking

Bob wants to ensure he arrives at the airport on time. He opens the ROAd application and books a robotaxi in advance, specifying his home as the pickup location, the airport as the destination, and scheduling the pickup for 7:00 AM the following day. The system applies the active allocation strategy, selects a suitable robotaxi and atomically reserves its availability for the required time interval, avoiding conflicts with other immediate rides and advance bookings. The next morning, shortly before the scheduled time, the system activates the reservation. The selected robotaxi transitions from available to assigned and then to arriving, and Bob is notified of the ride progress.

Scenario 3: Automated and Manual Allocation Strategy Management

Mark is a Fleet Operator supervising the city's robotaxi fleet through the ROAd web interface while the system is in Auto Mode. During a period of heavy rain, traffic in the city increases. The system detects the traffic level as Medium and displays a notification on Mark's dashboard suggesting a switch from the default Nearest Available Vehicle strategy to the Minimum Estimated Time of Arrival (ETA) policy to mitigate delays. Mark observes the situation but decides to remain in Auto Mode. As congestion continues to rise, the system detects a High traffic level and automatically switches the active strategy to Minimum ETA to ensure faster passenger pickups. Later, Mark determines that a specific operational condition requires a manual override, he manually selects the Nearest Available

strategy. The system immediately transitions to Manual Mode, suspending all automated strategy changes until Mark explicitly re-enables Auto Mode via the dashboard interface.

Scenario 4: Dynamic Fleet Rebalancing

The ROAd system periodically estimates zone demand using historical demand data and information about relevant active events. Late in the evening, a major concert ends at the city stadium. The system predicts this area as a high-demand zone. To minimize customer wait times, the system proactively initiates dynamic fleet rebalancing. It identifies several idle robotaxis scattered in low-demand areas of the city and triggers their repositioning toward the stadium. Mark, the Fleet Operator, observes these rebalancing movements on his monitor as the vehicles are routed to efficiently manage the increases in ride requests.

2.2 Domain Model

The domain model captures the core entities of the ROAd ecosystem and the relationships between them, representing the conceptual structure of the urban autonomous mobility service.

2.2.1 Main Entities

- ▶ **User:** An abstract entity representing any person interacting with the system, holding account credentials and personal data.
- ▶ **Passenger:** A specialized user who utilizes the mobile-friendly web application to request or book rides and receive notifications.
- ▶ **Fleet Operator:** A specialized user responsible for supervising the fleet, selecting allocation strategies, and monitoring rebalancing through the web interface.
- ▶ **Robotaxi:** An autonomous vehicle characterized by its unique ID, current GPS position, and operational state (Available, Assigned, Arriving, Arrived, In Ride, Maintenance).
- ▶ **Ride Request:** A conceptual entity representing a passenger's need for transport. It contains the pickup location, destination, and status. A request is initially pending, then it can be accepted if a suitable robotaxi is assigned, rejected if no feasible vehicle is available, cancelled by the passenger, or completed after the ride has ended.
 - ▶ **Immediate Ride:** A request to be served as soon as possible.
 - ▶ **Advance Booking:** A request scheduled for a specific future date and time. When accepted, it reserves the selected Robotaxi for the corresponding time interval.
- ▶ **Ride:** An actual transport service generated from an accepted Ride Request. It represents the execution of the trip assigned to a Robotaxi, from pickup to destination, and keeps track of its lifecycle through states such as Scheduled, Waiting for Pickup, In Progress, Completed, or Cancelled.
- ▶ **Location:** A geographical point defined by coordinates, used to specify Pickup and Destination points.
- ▶ **Zone:** A partition of the urban area used to track demand and trigger fleet rebalancing.
- ▶ **System Mode:** An attribute of the ROAd system that determines the control logic for vehicle allocation, supporting Auto and Manual modes.

- ▶ **Traffic Level:** A dynamic parameter provided by external Traffic Information Source, classified as Low, Medium, or High, which serves as the input for automated strategy switching.
- ▶ **Allocation Strategy:** The policy used for vehicle assignment.
 - ▶ **Nearest Available Vehicle:** Selects the eligible robotaxi geographically closest to the pickup location (default).
 - ▶ **Minimum ETA:** Selects the robotaxi with the shortest ETA to minimize passenger wait times during congestion.
- ▶ **Maintenance Event:** An event representing a period in which a Robotaxi is unavailable due to inspection, repair, cleaning, or any other operational reason. During a maintenance event, the corresponding Robotaxi cannot be assigned to ride requests.

2.2.2 Key Relationships

- ▶ **Passenger - Ride Request:** A Passenger can submit multiple Ride Requests over time, but each request is associated with exactly one Passenger.
- ▶ **Ride Request - Location:** Each Ride Request must specify exactly one Pickup Location and one Destination.
- ▶ **Passenger - Ride:** A Passenger can take multiple Rides over time, but each Ride is associated with exactly one Passenger.
- ▶ **Robotaxi - Ride Request:** A Robotaxi is selected for a Ride Request by the system based on the active strategy. A Robotaxi may serve multiple requests over time, but its assignments and reservations must never overlap.
- ▶ **Fleet Operator - Allocation Strategy:** The Fleet Operator selects and can change the active Allocation Strategy at runtime.
- ▶ **Fleet Operator - System Mode:** The Fleet Operator can switch the system between Auto and Manual modes via the web interface.
- ▶ **System - Allocation Strategy:** In Auto Mode, the system dynamically selects the Allocation Strategy based on the current Traffic Level and provides suggestions to the operator.
- ▶ **Fleet Operator - Allocation Strategy:** In Manual Mode, the Operator directly selects the strategy; performing this action while in Auto Mode triggers an automatic override and switches the system to Manual Mode.
- ▶ **Traffic Information Source - Traffic Level:** The external service provides real-time Traffic Level data used by the system to evaluate strategy transitions.
- ▶ **Robotaxi - Zone:** Every Robotaxi is located within a specific Zone at any given time. Zones are used to identify High-Demand areas for Rebalancing.
- ▶ **Fleet - Robotaxi:** A Fleet is composed of multiple Robotaxis managed by the system.
- ▶ **Ride Request - Ride:** An accepted Ride Request generates one Ride. Requests rejected or cancelled before acceptance do not generate a Ride. Cancellation after acceptance changes the associated Ride to Cancelled.
- ▶ **Robotaxi - Maintenance Event:** A Robotaxi can be associated with multiple Maintenance Events over time. When a Maintenance Event is active, the Robotaxi is considered unavailable for assignment.

2.2.3 Class Diagrams

The diagrams presented in this section describe the domain model of the ROAd system. They provide a **conceptual view** of the main entities involved in the autonomous robotaxi allocation service, together with their relevant attributes and relationships.

Since the purpose of these diagrams is to support requirements analysis, they abstract from implementation details and do not aim to represent the complete software architecture or all the classes that will be introduced in the design phase.

Note: Some attributes are represented through enumerations in order to make the possible states or categories explicit.

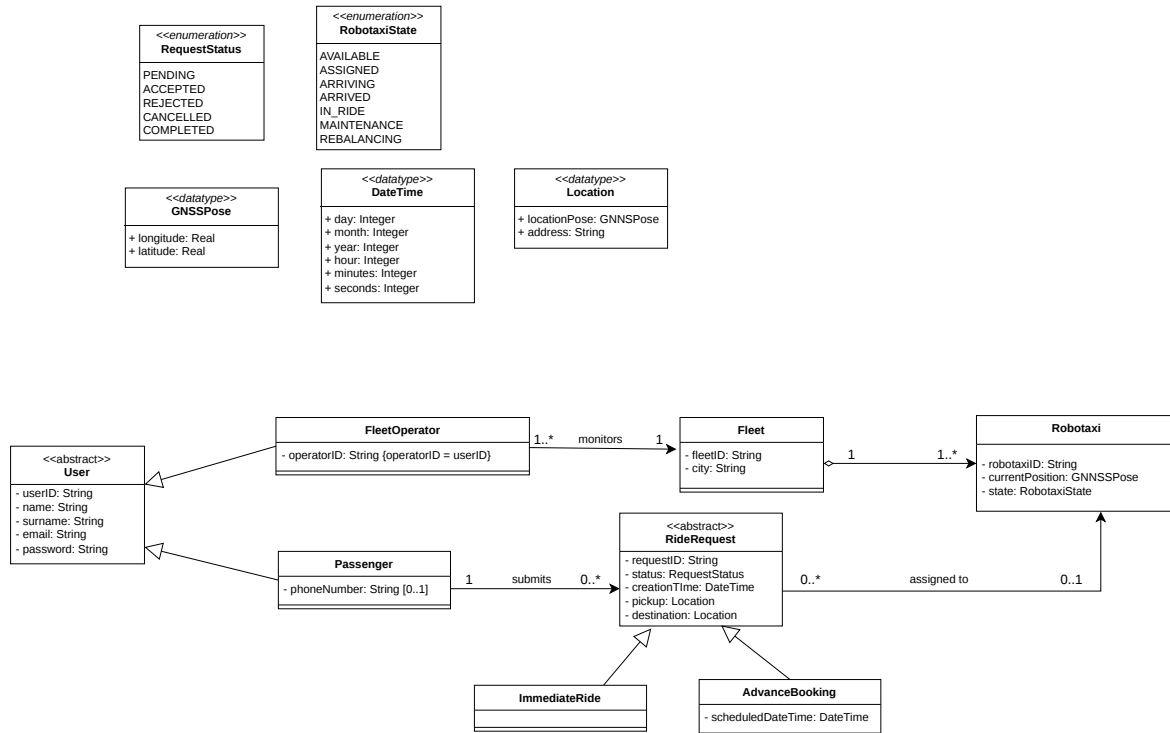


Figure 2.1: Core ROAd domain entities

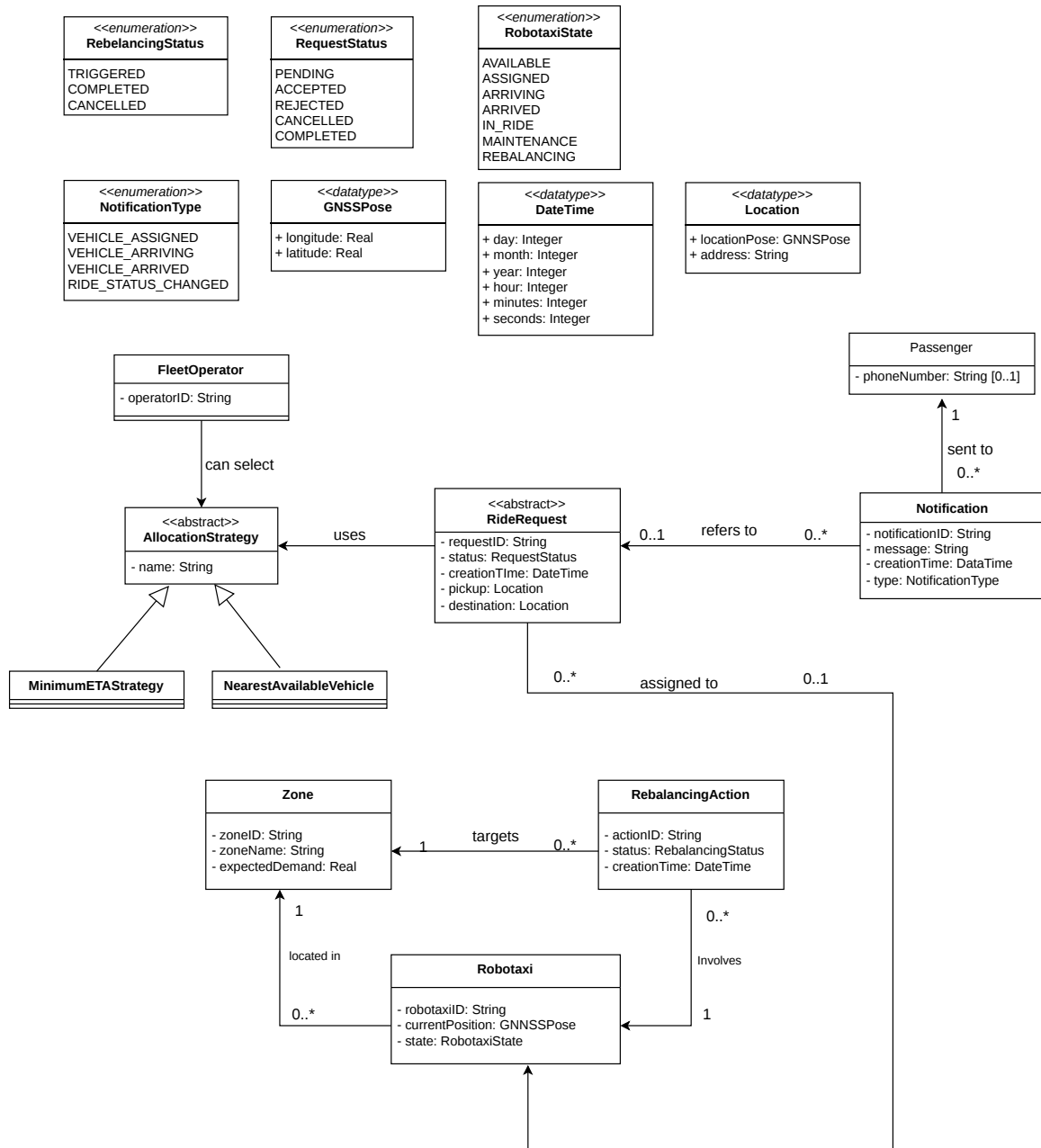


Figure 2.2: Allocation, rebalancing, and notification entities

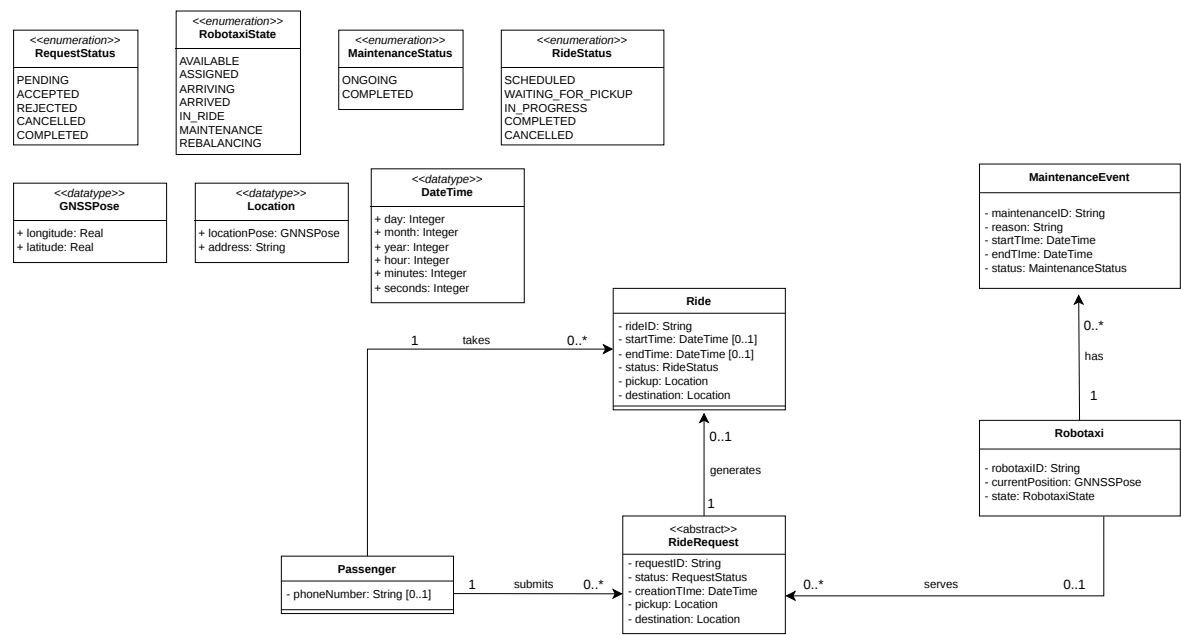


Figure 2.3: Ride execution and robotaxi lifecycle entities

2.3 User Characteristics

The ROAd system is designed to be used by two primary categories of users, each with distinct roles, technical requirements, and operational goals. Understanding these characteristics is essential to ensure the system meets the practical needs of urban mobility.

Passengers

Passengers represent the general public and are the primary consumers of the robotaxi service. Their main objective is to obtain reliable, point-to-point transportation with minimal effort and waiting time.

- ▶ **Technical Literacy:** Passengers are expected to have a basic to intermediate level of familiarity with mobile-friendly web applications. The interface must be intuitive, requiring no specialized training.
- ▶ **Needs:** They require a streamlined process for requesting immediate rides or scheduling advance bookings. Real-time updates and clear notifications regarding vehicle status (e.g., assignment, arrival) are critical for a positive user experience.
- ▶ **Mobility Patterns:** Their usage varies from daily commuting to occasional trips (e.g., airport transfers), necessitating a system that can handle both spontaneous and planned requests efficiently.

Fleet Operators

Fleet Operators are professional users responsible for the supervision, management, and optimization of the autonomous vehicle fleet. They act as the “controllers” of the system’s high-level logic.

- ▶ **Technical Literacy:** They are expected to be proficient in using web-based management interfaces and interpreting data visualizations (e.g., fleet maps). They must understand the implications of different vehicle allocation strategies.
- ▶ **Responsibilities:** Their primary tasks include monitoring the real-time distribution and state of the fleet and overseeing proactive rebalancing operations. Crucially, they act as supervisors of the automated allocation logic; they are responsible for evaluating system-generated suggestions triggered by traffic fluctuations and executing manual overrides to take direct control of the active allocation strategy during exceptional or unforeseen operational conditions.
- ▶ **Needs:** They require a comprehensive “command and control” dashboard that provides a clear overview of the traffic level in the service area, information about rebalancing actions, and vehicle lifecycle transitions. Furthermore, the interface must clearly display the current operational state of the system (Auto or Manual mode) and provide immediate alerts for strategy switch suggestions, empowering them to make informed decisions safely and without service interruption.

2.4 Product Functions

The main functions of the ROAd system are categorized based on the primary interactions between the software and its users (Passengers and Fleet Operators) and its autonomous management logic. These functions are designed to satisfy the project goals.

User Management and Authentication

- ▶ **[R1] Registration and Login:** The system shall provide account access through Passenger registration and authentication of Passengers and Fleet Operators into their respective interfaces.
- ▶ **[R2] Profile Management:** The system shall allow authenticated Users to update their personal information and credentials.

Ride Management

- ▶ **[R3] Immediate Ride Request:** The system shall allow Passengers to request a robotaxi for immediate pickup by specifying a pickup location and a destination.
- ▶ **[R4] Advance Booking:** The system shall allow Passengers to book a ride for a future date and time, specifying pickup and destination locations. Confirmation shall depend on the availability of a suitable robotaxi whose required time interval can be reserved without conflicting with existing commitments.
- ▶ **[R5] Automated Vehicle Allocation:** The system shall automatically assign an eligible robotaxi to a ride request according to the globally active allocation strategy, taking into account the vehicle's operational state and existing reservations.
- ▶ **[R6] Ride Status Notifications:** The system shall notify Passengers about the progress of their rides, including vehicle assignment, estimated arrival time, arrival at the pickup point, and ride completion.
- ▶ **[R14] Ride Cancellation:** The system shall allow a Passenger to cancel an immediate ride request or an advance booking before the ride begins, including while the assigned robotaxi is approaching or waiting at the pickup point. Cancellation shall release any associated reservation and pickup assignment, return an assigned vehicle to the available state, and make the released time interval available for subsequent bookings. Cancellation shall be rejected once the ride has begun.

Fleet Supervision and Control

- ▶ **[R7] Fleet Monitoring:** The system shall provide Fleet Operators with an overview of the position and operational state of every robotaxi, updated as new vehicle data becomes available.
- ▶ **[R8] Runtime Strategy Selection:** The system shall allow Fleet Operators to change the globally active allocation strategy between Nearest Available Vehicle and Minimum ETA at runtime without interrupting the service.
- ▶ **[R9] Maintenance Management:** The system shall allow Fleet Operators to manage robotaxi maintenance by placing eligible vehicles in maintenance and returning them to service when the intervention is completed. Vehicles in maintenance shall be excluded from ride allocation.

Optimization and Rebalancing

- ▶ **[R10] Demand Analysis:** The system shall estimate demand for each service-area zone using historical demand data and information about relevant active events, identifying zones with higher expected demand.
- ▶ **[R11] Proactive Fleet Rebalancing:** The system shall automatically initiate the repositioning of idle robotaxis from zones with surplus availability toward zones where expected demand exceeds

vehicle availability, when suitable vehicles can be repositioned without compromising existing commitments.

Dynamic Allocation Strategy Management

Auto and Manual are the two modes governing the globally active allocation strategy. They do not determine whether fleet rebalancing is enabled.

- ▶ **[R12] Auto Mode (Default):** In Auto Mode, the system shall determine the active allocation strategy from the current service-area traffic level, using Nearest Available Vehicle as the initial default. At Medium, the system shall alert the Fleet Operator and suggest Minimum ETA while retaining the active strategy. At High, the system shall select Minimum ETA. At Low, the system shall select Nearest Available Vehicle. The stability constraint in NFR9 shall apply.
- ▶ **[R13] Manual Mode (Override):** An Operator's manual strategy selection shall place the system in Manual Mode, in which the selected strategy remains active regardless of traffic changes. This mode shall persist until the Operator explicitly re-enables Auto Mode. Reactivation shall immediately apply the current traffic-level rules in R12: Nearest Available Vehicle at Low, Minimum ETA at High, and retention of the active strategy at Medium.

2.5 Quality Requirements and System Constraints

This section groups quality requirements, system-wide consistency constraints, and architectural quality objectives. The existing NFR identifiers are retained for traceability; they do not imply that every statement below describes an externally measurable nonfunctional quality.

Concurrency and Responsiveness

- ▶ **[NFR1] Concurrency Handling:** The system shall support concurrent ride requests and vehicle-data updates from multiple users and robotaxis while preserving the consistency constraints specified in NFR4 and NFR5.
- ▶ **[NFR2] Automatic Update Delivery:** The system shall deliver relevant state-change updates to connected Passenger and Fleet Operator interfaces without requiring a manual refresh.

System-wide Consistency Constraints

- ▶ **[NFR4] Data Consistency:** Vehicle assignments and reservations shall remain non-overlapping for each robotaxi, including when conflicting requests are processed concurrently.
- ▶ **[NFR5] State Consistency:** Every robotaxi state change shall conform to the lifecycle model. An operation requesting an inadmissible transition shall be rejected without applying that transition.
- ▶ **[NFR9] Stability of Auto-Switching:** After an automatic switch to Minimum ETA at High traffic, a subsequent Medium traffic level shall not cause a return to Nearest Available Vehicle. Automatic reversion shall occur only at Low, as specified in R12.
- ▶ **[NFR10] Consistency of Manual Override:** An Operator's manual strategy selection and the corresponding transition to Manual Mode shall take effect atomically. A concurrent automatic strategy evaluation shall not overwrite that selection once Manual Mode has taken effect.

Usability Constraints

- ▶ **[NFR6] Interface Clarity:** The Passenger and Fleet Operator interfaces shall consistently distinguish ride states, vehicle states, allocation strategies, and operational modes, and shall provide explicit feedback indicating whether a user action succeeded or was rejected.

Architectural Quality Objectives

- ▶ **[NFR3] Scalability:** The system architecture shall permit the number of registered Passengers and managed robotaxis to increase without changes to domain logic.
- ▶ **[NFR7] Extensibility of Allocation Logic:** The allocation logic shall allow new allocation strategies to be added without modifying the core ride-request management logic.
- ▶ **[NFR8] Separation of Concerns:** Account management, ride management, fleet management, and external-service integration shall have distinct responsibilities, so that a change to an external-service integration does not require changes to unrelated domain logic.

2.6 Assumptions, Dependencies and Constraints

The following domain assumptions, external dependencies, and constraints define the operating environment and scope of the ROAd system.

Domain Assumptions

- ▶ **[D1] Vehicle Autonomy:** Robotaxis are capable of autonomous driving and safe navigation within the service area. Their onboard systems execute navigation commands and provide position and operational-state information to ROAd.
- ▶ **[D2] Connectivity:** Robotaxis and Users have network connectivity sufficient to exchange position updates, state changes, ride requests, and notifications with ROAd during normal operation.
- ▶ **[D3] User Behavior:** Passengers are present at the specified pickup location when their assigned robotaxi arrives.

Dependencies

- ▶ **Mapping and Routing Services:** The system relies on external geographical and routing information to locate pickup and destination points, compute routes and distances, and estimate travel times.
- ▶ **Traffic Information:** Automatic allocation-strategy management relies on a source providing the current service-area traffic level, classified as Low, Medium, or High. This information need not be supplied by the same provider used for mapping and routing.
- ▶ **Demand Information:** Demand estimation relies on historical demand data for the service-area zones and information about relevant active events. These inputs support the identification of zones requiring additional vehicle availability.
- ▶ **Vehicle Communication:** Fleet monitoring and ride execution rely on the exchange of navigation commands and vehicle telemetry with the robotaxis' onboard systems.

Constraints

- ▶ **[C1] Non-Overlapping Assignments:** A robotaxi shall not have assignments or reservations with overlapping time intervals. This invariant shall also hold when requests are processed concurrently, as specified in NFR4.
- ▶ **[C2] Scope Limitation:** ROAd covers account management, ride requests and bookings, vehicle allocation, fleet monitoring, maintenance management, passenger notifications, and automatic fleet rebalancing. Route computation is delegated to external services. Physical vehicle control, onboard sensor processing, and payment processing are outside the project scope.

Additional Models

3.1 Requirements-level Sequence Diagrams

The sequence diagrams presented in this section describe the main interactions between external actors and the ROAd system at requirements level. They abstract from internal software components and focus on the order of events needed to satisfy the main functionalities of the system.

Immediate Ride Request

The Passenger submits pickup and destination locations. The ROAd system validates the request and evaluates eligible robotaxis according to the active allocation strategy and existing commitments, using external routing information where needed. If a suitable robotaxi can be assigned, the system confirms the assignment, directs the vehicle to the pickup location, and notifies the Passenger about its estimated arrival time and subsequent arrival. If the request is invalid or no suitable robotaxi can be assigned, the system rejects the request and informs the Passenger.

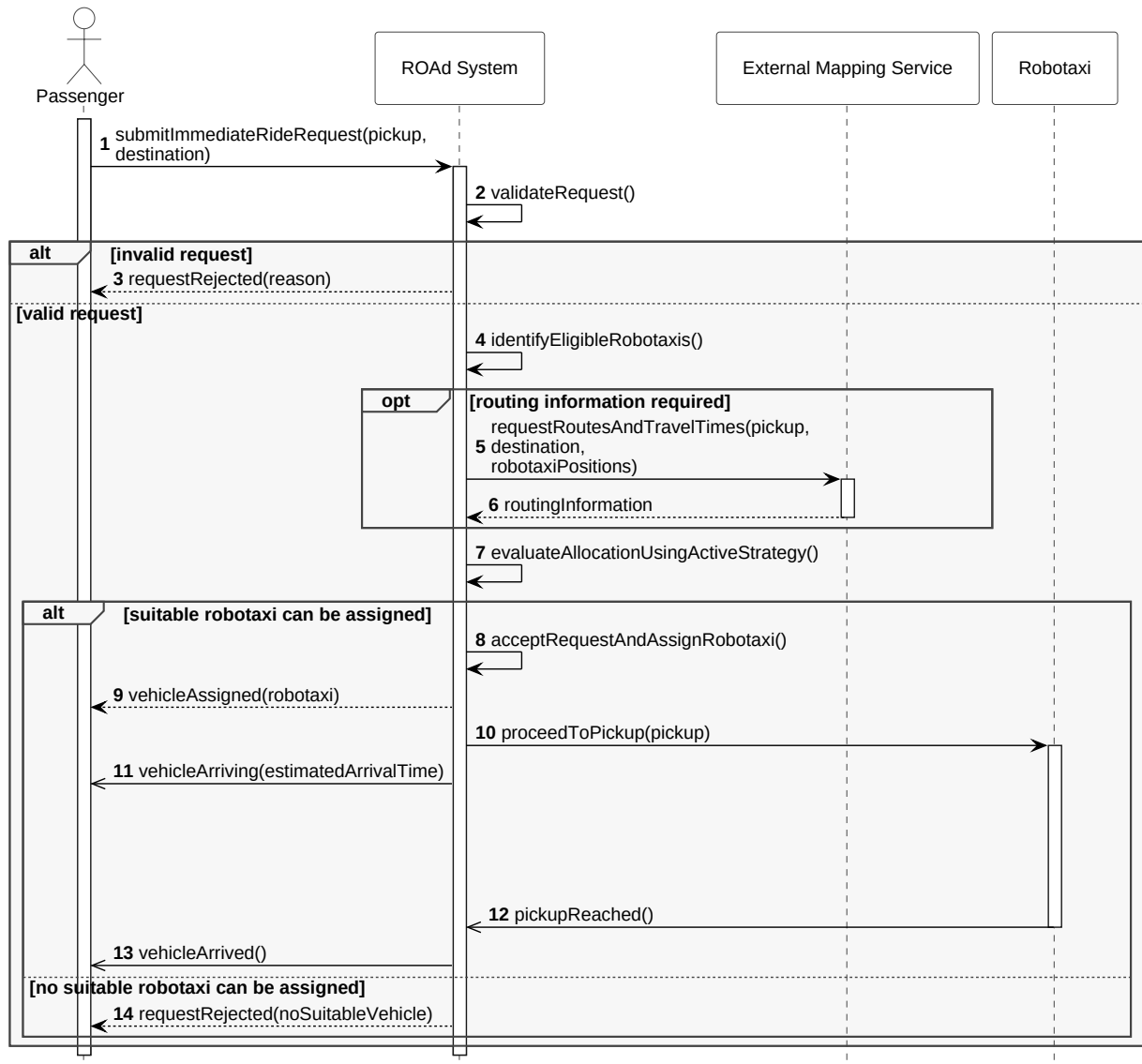


Figure 3.1: Requirements-level sequence diagram for immediate ride request

Advance Ride Booking

The Passenger submits pickup and destination locations and a future pickup date and time. The ROAd system validates the request and evaluates vehicle availability for the required time interval, using external routing information and the active allocation strategy. If a suitable robotaxi can be reserved without conflicting with existing commitments, the system records the booking and confirms it to the Passenger. Otherwise, the request is rejected and the Passenger is informed.

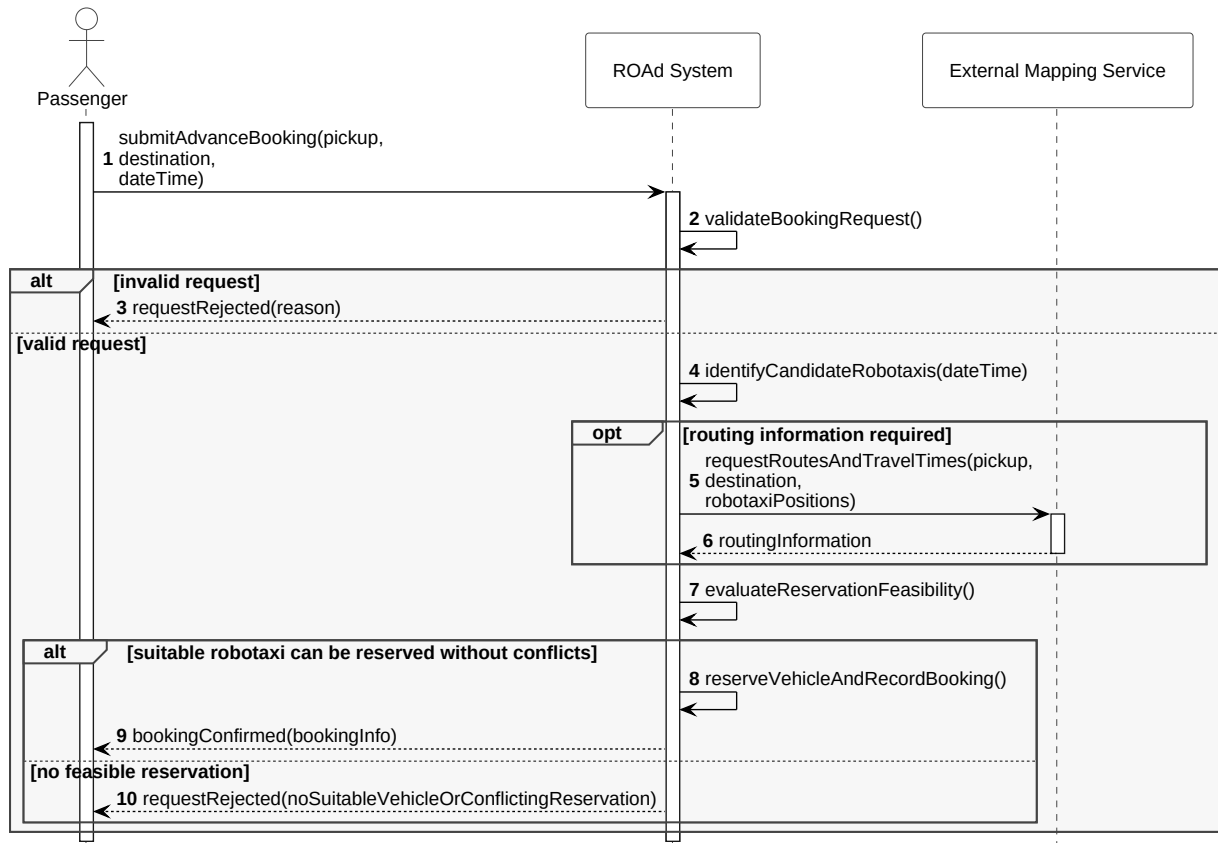


Figure 3.2: Requirements-level sequence diagram for advance ride booking

Fleet Rebalancing

The ROAd system periodically estimates demand for each zone using historical demand data and relevant active events, and compares it with the available vehicle supply. When an imbalance exists, the system selects eligible idle robotaxis from zones with surplus availability and automatically directs them toward zones with high demand, while preserving existing commitments. The Fleet Operator is notified of each initiated repositioning. If no imbalance exists or no suitable vehicle can be repositioned, no new rebalancing action is initiated.

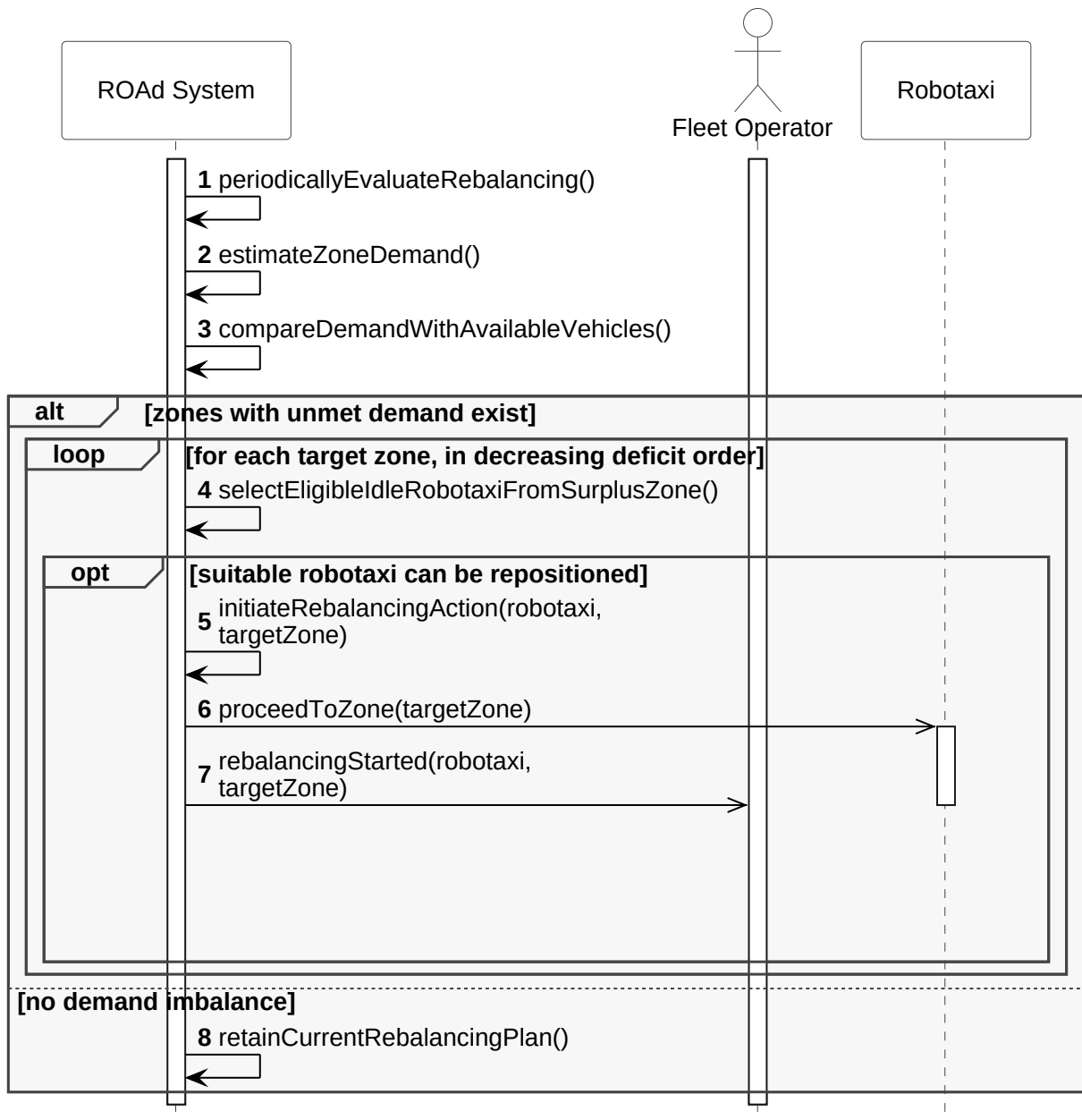


Figure 3.3: Requirements-level sequence diagram for fleet rebalancing

3.2 Finite State Machines

The finite state machine presented in this section describes the robotaxi lifecycle at requirements level. It defines the admissible operational-state transitions associated with ride assignment, pickup, ride execution, rebalancing, cancellation, and maintenance.

Robotaxi State Machine

A robotaxi can be assigned to a ride request when it is in the AVAILABLE or REBALANCING state, provided that the assignment does not conflict with existing commitments. Assigning a ride during rebalancing interrupts the repositioning activity.

An assigned robotaxi moves toward the pickup location, waits for the Passenger, and executes the ride once the Passenger boards. Upon ride completion, it returns to AVAILABLE. If the ride is cancelled

before it begins, the pickup assignment is terminated and the robotaxi returns to AVAILABLE, including when it is already approaching or waiting at the pickup location.

An available robotaxi may also begin automatic rebalancing or enter maintenance. Rebalancing ends when the target zone is reached or is interrupted by a ride assignment. A robotaxi in MAINTENANCE cannot be assigned to rides and returns to AVAILABLE when maintenance is completed.

Ride cancellation is not permitted from IN_RIDE. Cancelling an advance booking that has not yet resulted in an operational vehicle assignment releases its reservation without requiring a robotaxi state transition.

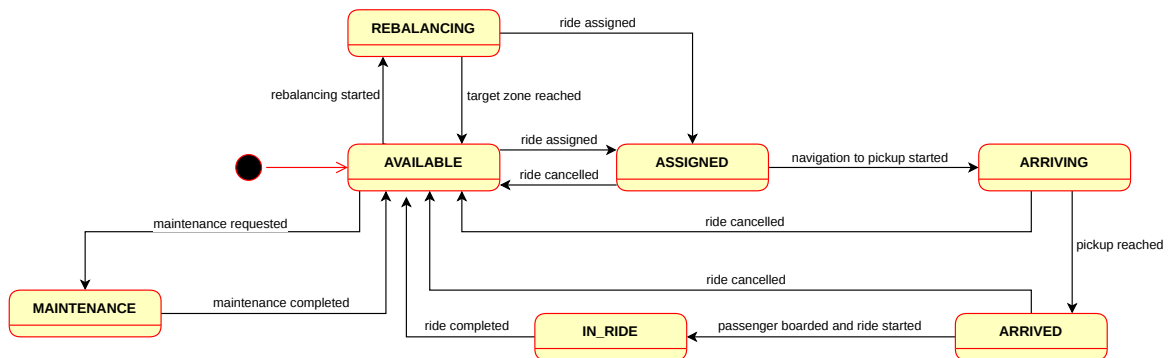


Figure 3.4: Requirements-level finite state machine of the robotaxi lifecycle

References

- ▶ ROAD: Robotaxi Optimized Allocation project proposal
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